Table 17: Prompt used to generate experiment

```
You are a scientific expert tasked with designing rigorous,
feasible experiments based on specified scientific questions
and the methodologies derived from the idea I provide, along
with relevant past research.  Your goal is to assist researchers
in systematically testing hypotheses and validating innovative
discoveries that could significantly advance their fields.

Past Related Research Experiments:  [Past experiments]
Here are the entities you need to know:  [Entities]
Here is the idea you need to design an experiment for:  [Idea]
Please propose a detailed experimental plan addressing the
following points:
1.  Experimental Design:  Develop rigorous experiments to
ensure the reliability and validity of your results.  Provide
a comprehensive explanation of the baseline used, comparative
methods, ablation study design, and criteria for data analysis
and result evaluation.  Clarify how these components collectively
reinforce and validate the conclusions of your research.  Structure
your experimental design in a clear, logical, and step-by-step
manner, ensuring each step is well-defined and easy to understand.
2.  Implementation of Technologies/Methods:  If your experimental
design involves specific technologies or methodologies, describe
the implementation process in detail, including key technical
aspects.  For any critical concepts utilized, provide thorough
explanations.  For instance, if you propose a modular approach,
detail its construction, components, and functionality.
3.  Feasibility Assessment:  Ensure your experimental plan is
realistic, considering technological availability, timelines,
resources, and personnel.  Identify potential challenges and
propose strategies for addressing them.
4.  References to Previous Studies:  When citing related
literature, include titles and pertinent details of the original
papers.  Strive to use as many references as necessary to support
your experimental design.
5.  Visual Aids:  If useful, provide pseudo code or a flowchart to
illustrate the implementation process.  For example, you can use
pseudo code to detail the core algorithm or the model architecture,
or employ a flowchart to map out the experimental procedure and
data flow.
6.  Clarity of Language:  Use straightforward language to describe
your methods, assuming the reader may have limited knowledge of
the subject matter.  Avoid complex jargon and utilize accessible
terminology.  If professional terms are necessary, please provide
clear and detailed explanations.

Please output strictly in the following format:
Experiment:
Step1:  ...
Step2:  ...
...
```

Table 18: Prompt used to review experiment

```
You are an expert in paper review. Your task is to analyze whether
a given experiment can effectively verify a specific idea, as well
as assess the detail and feasibility of the experiment.

Here are the related entities you need to know: [Entities]
The idea presented is: [Idea]
The corresponding experiment designed for this idea is:
[Experiment]
Please conduct your analysis based on the following criteria:
1. Can the experiment validate the idea? If not, identify
the issues and suggest improvements to enhance its verification
capability and feasibility.
2. Are there specific experimental procedures that are confusing
or poorly designed? Discuss any methods that may not be feasible,
uncertainties in constructing the dataset, or a lack of explanation
regarding the implementation of certain methods.
3. Evaluate the clarity, detail, reasonableness, and feasibility
of the experimental design.
4. Provide suggestions for improving the experiment based on the
shortcomings identified in your analysis.
5. Focus solely on the experiment design; please refrain from
altering the original idea.
6. Ensure that your suggestions are constructive, concise, and
specific.

Please strictly follow the following format for output:
Suggestion: ...
```

Table 19: Prompt used to get query for search paper to refine experiment

```
You are a research expert tasked with refining and improving an
experimental plan based on the feedback received.

The experimental plan you proposed is as follows: [Experiment]
You have received the following suggestions for improvement:
[Suggestions]
Please decide whether you need to search for relevant papers to
obtain relevant knowledge to improve your experiment.
If you need to search for relevant papers, please provide a search
query for literature search, else provide "".
For example: if suggestions say that the dynamic query additional
information and update knowledge graph described in the experiment
is not clearly described, so you need to output "dynamic knowledge
graph update".

Please output strictly in the following format:
Query:...
```

Table 20: Prompt used to refine experiment

```
You are a research expert tasked with refining and improving an
experimental plan based on the feedback received.

The information of the literature you maybe need to refer to are
as follows:  [Searched paper information]
The experimental plan you proposed is as follows:  [Experiment]
Please propose a detailed experimental plan addressing the
following points:
1.  Experimental Design:  Develop rigorous experiments to
ensure the reliability and validity of your results.  Provide
a comprehensive explanation of the baseline used, comparative
methods, ablation study design, and criteria for data analysis
and result evaluation.  Clarify how these components collectively
reinforce and validate the conclusions of your research.  Structure
your experimental design in a clear, logical, and step-by-step
manner, ensuring each step is well-defined and easy to understand.
2.  Implementation of Technologies/Methods:  If your experimental
design involves specific technologies or methodologies, describe
the implementation process in detail, including key technical
aspects.  For any critical concepts utilized, provide thorough
explanations.  For instance, if you propose a modular approach,
detail its construction, components, and functionality.
3.  Feasibility Assessment:  Ensure your experimental plan is
realistic, considering technological availability, timelines,
resources, and personnel.  Identify potential challenges and
propose strategies for addressing them.
4.  References to Previous Studies:  When citing related
literature, include titles and pertinent details of the original
papers.  Strive to use as many references as necessary to support
your experimental design.
5.  Visual Aids:  If useful, provide pseudo code or a flowchart to
illustrate the implementation process.  For example, you can use
pseudo code to detail the core algorithm or the model architecture,
or employ a flowchart to map out the experimental procedure and
data flow.
6.  Clarity of Language:  Use straightforward language to describe
your methods, assuming the reader may have limited knowledge of
the subject matter.  Avoid complex jargon and utilize accessible
terminology.  If professional terms are necessary, please provide
clear and detailed explanations.
You have received the following suggestions for
improvement:[Suggestions]
Please refine your experimental plan based on the feedback
provided.  Ensure your refined plan is feasible, clearly defined,
and addresses the feedback you received.

Please output strictly in the following format:
Experiment:  ...
```